

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT: Artificial intelligence

PRACTICAL NO. 9

Aim: Aprior

CODE:

```
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules

data = {
    "TransactionID": [
        1,
        1,
        1,
        2,
        2,
        3,
        3,
        3,
        4,
        4,
        5,
        5,
        5,
        6,
        6,
        7,
        7,
        7,
        8,
        8,
    ],
    "Item": [
        "Coffee",
        "Pastry",
        "Toast", # Order 1
        "Coffee",
        "Pastry", # Order 2
        "Tea",
        "Cookies",
        "Juice", # Order 3
    ]
}
```

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```
        "Coffee",
        "Pastry", # Order 4
        "Coffee",
        "Toast",
        "Cookies", # Order 5
        "Tea",
        "Cookies", # Order 6
        "Coffee",
        "Pastry",
        "Cookies", # Order 7
        "Tea",
        "Toast", # Order 8
    ],
}
df = pd.DataFrame(data)

basket = pd.crosstab(df["TransactionID"], df["Item"]).astype(bool)

frequent_itemsets = apriori(basket, min_support=0.25, use_colnames=True)
frequent_itemsets = frequent_itemsets.sort_values(
    by="support", ascending=False
).reset_index(drop=True)

print("--- Top Frequent Itemsets ---")
print(frequent_itemsets)

rules = association_rules(
    frequent_itemsets, metric="confidence", min_threshold=0.60
)

clean_rules = rules[
    ["antecedents", "consequents", "support", "confidence", "lift"]
]
clean_rules = clean_rules.sort_values(by="lift",
ascending=False).reset_index(
    drop=True
)
```

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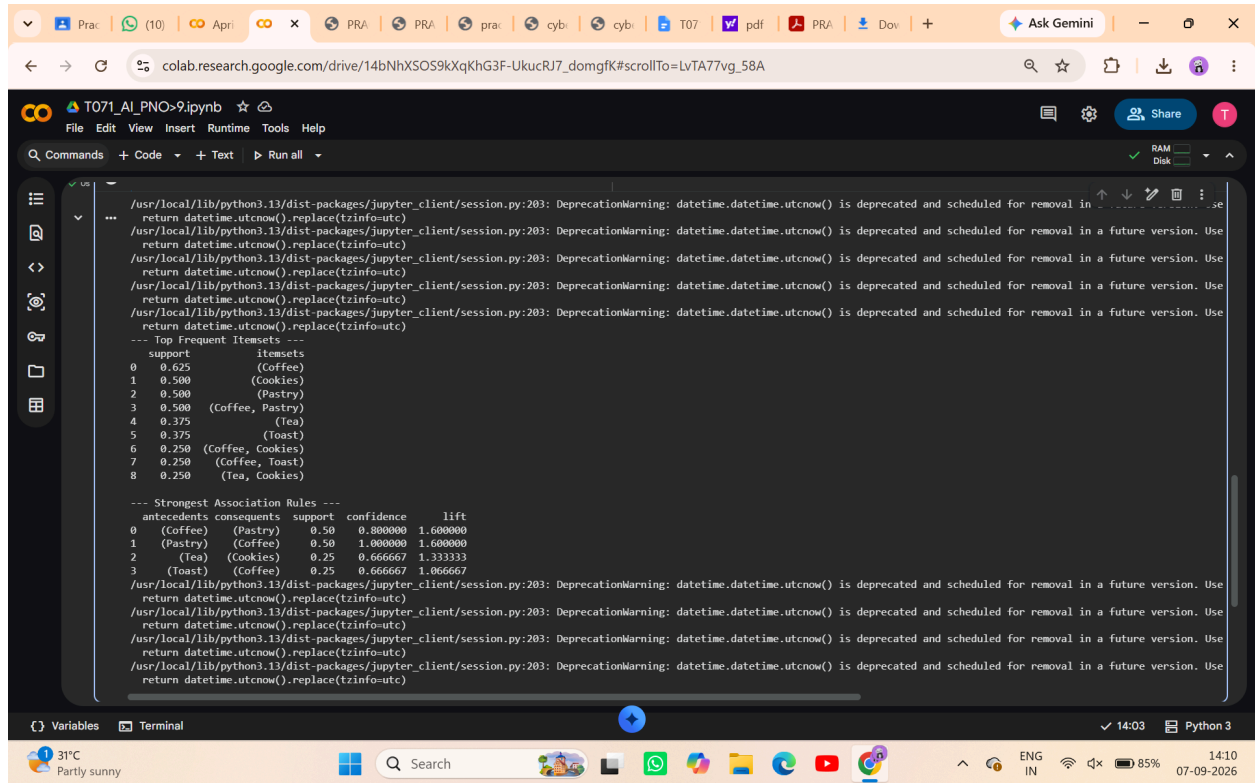
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```
print("\n--- Strongest Association Rules ---")
print(clean_rules)
```

OUTPUT:



The screenshot shows a Jupyter Notebook interface with a code cell containing the following output:

```
--- Top Frequent Itemsets ---
support  itemsets
0  0.625  (Coffee)
1  0.500  (Cookies)
2  0.500  (Pastry)
3  0.500  (Coffee, Pastry)
4  0.375  (Tea)
5  0.375  (Toast)
6  0.250  (Coffee, Cookies)
7  0.250  (Coffee, Toast)
8  0.250  (Tea, Cookies)

--- Strongest Association Rules ---
antecedents consequents support confidence lift
0  (Coffee) (Pastry) 0.50 0.800000 1.600000
1  (Pastry) (Coffee) 0.50 1.800000 1.600000
2  (Tea) (Cookies) 0.25 0.666667 1.333333
3  (Toast) (Coffee) 0.25 0.666667 1.066667
```

The notebook interface shows the file 'T071_AI_PNO>9.ipynb' and the output is displayed in the 'Text' tab. The bottom status bar indicates the system time is 14:03 and the Python version is 3.

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